Lab 2

Image registration

Important points

- Do not loop over pixels for things like matrix multiplication. Use vectorized functions such as

  np.dot, np.mul, @, et c.

- If you use LLMs, **you must** document your use (such as prompts and answers) **and be ready to submit them** and answer questions.

General reminders

- Ask questions in the discussion forum instead of individual emails to helps other to benefit from your questions. Can be done anonymously.
- Also, you are welcome to answer each others' questions. Again, it's anonymous.

Overview

- Model fitting
  - Implement least squares to fit a matrix transform
  - Implement RANSAC to handle noisy cases with outliers
- Image registration
  - Using ready-made functions:
    - Use SIFT to find features in images
    - Match features for one-to-one correspondences
    - Use RANSAC to find
- Image warping
  - Implement the image warping algorithm

Model fitting

- Implement least squares to fit a matrix transform of points
- Implement RANSAC to handle noisy cases with outlier point transforms

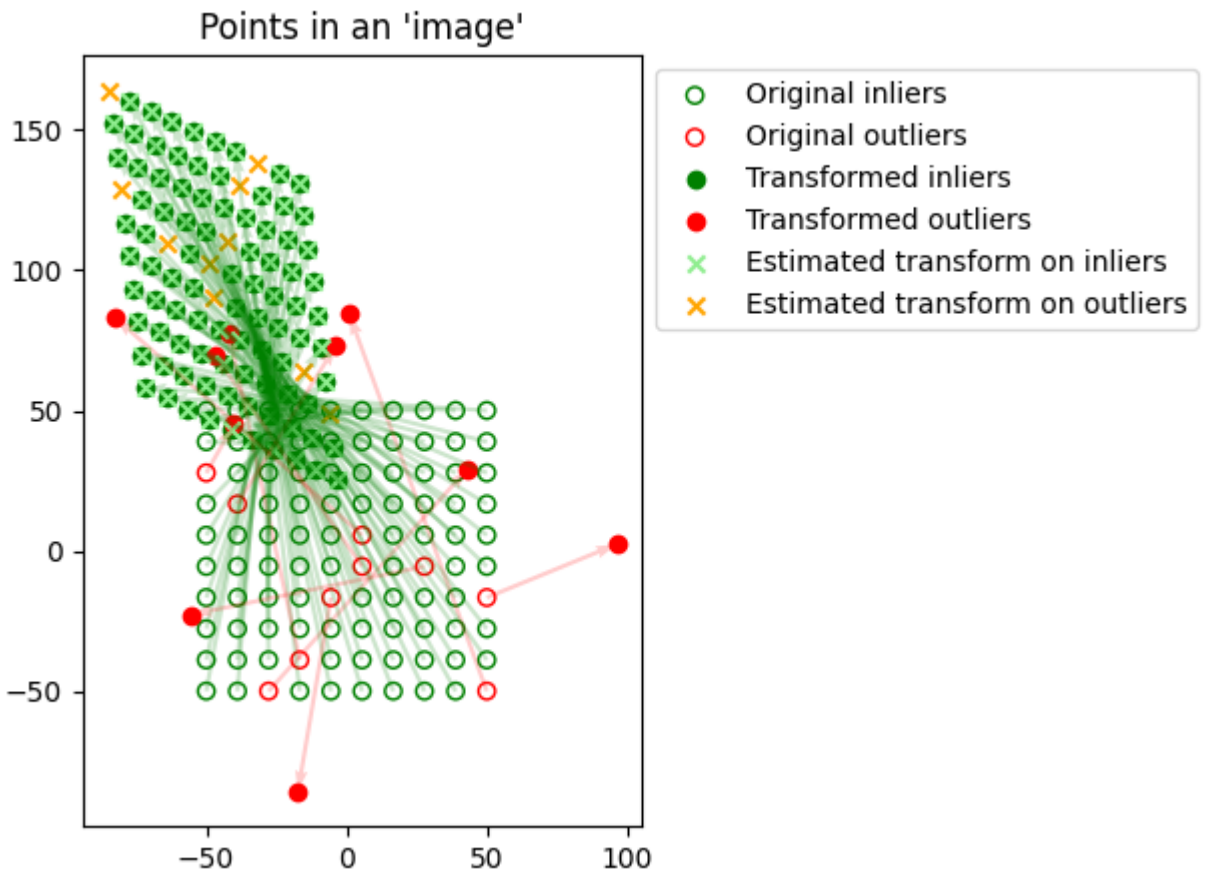


Image registration

1. Use SIFT to find features (points) in a source image and a target image. *
2. Match features for one-to-one correspondences *
3. Use RANSAC to find transform of the target to the source.
4. Warp the source to the target using the transform. *

* Using ready-made functions

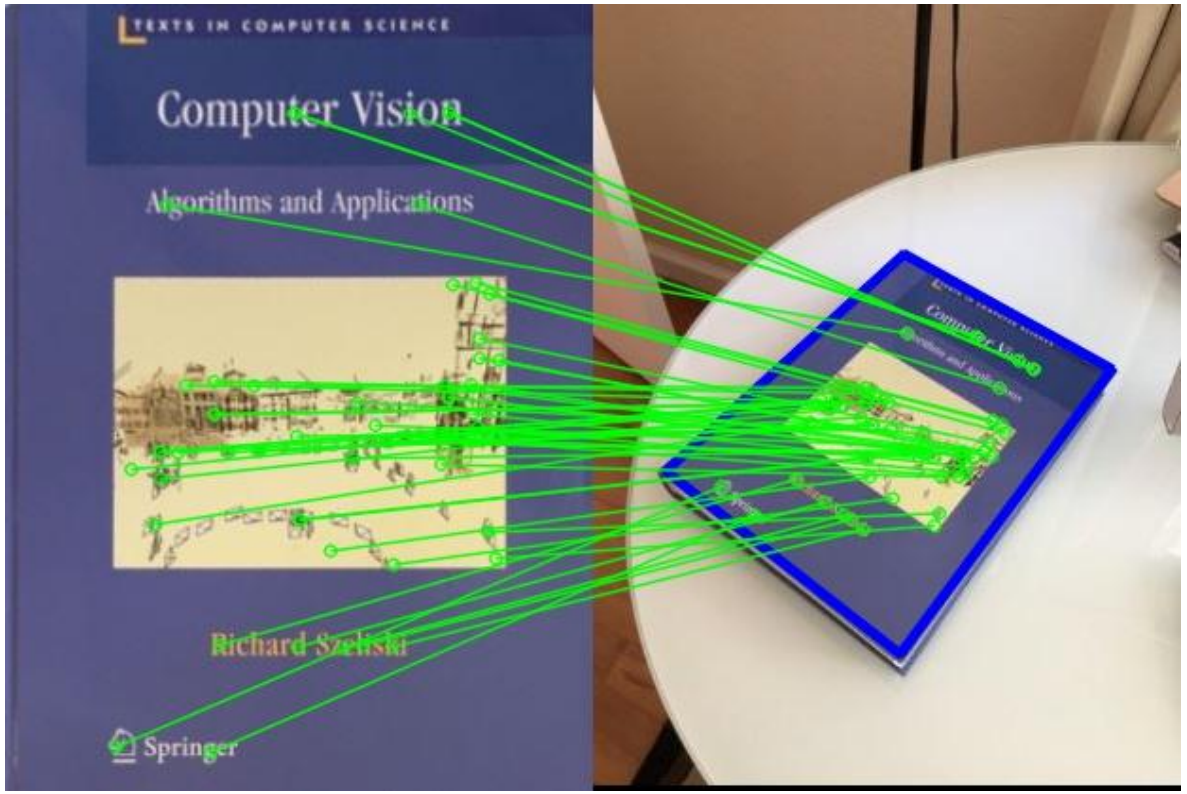


Image registration

1. Use SIFT to find features (points) in a source image and a target image. *
2. Match features for one-to-one correspondences *
3. **Use RANSAC to find transform of the target to the source.****
4. Warp the source to the target using the transform. *

* Using ready-made functions

** Source/moving/floating, target/fixed/reference

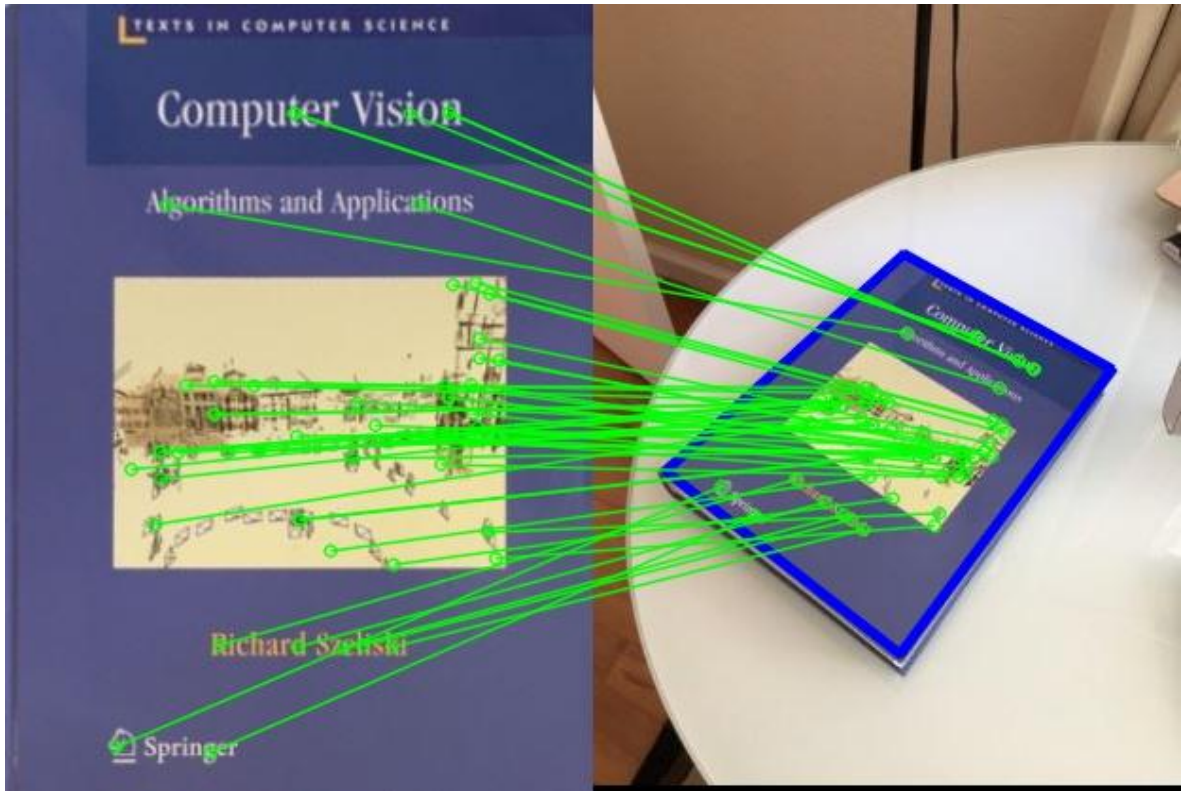
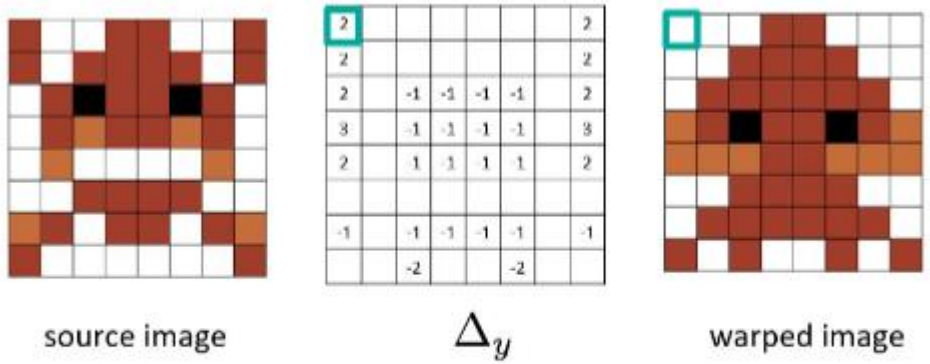


Image warping

- Transforming points is well and easy. What about the pixel intensities?

- Requires two steps:
  1. Transforming coordinates (e.g. affine transform)
  2. Sampling the pixel values

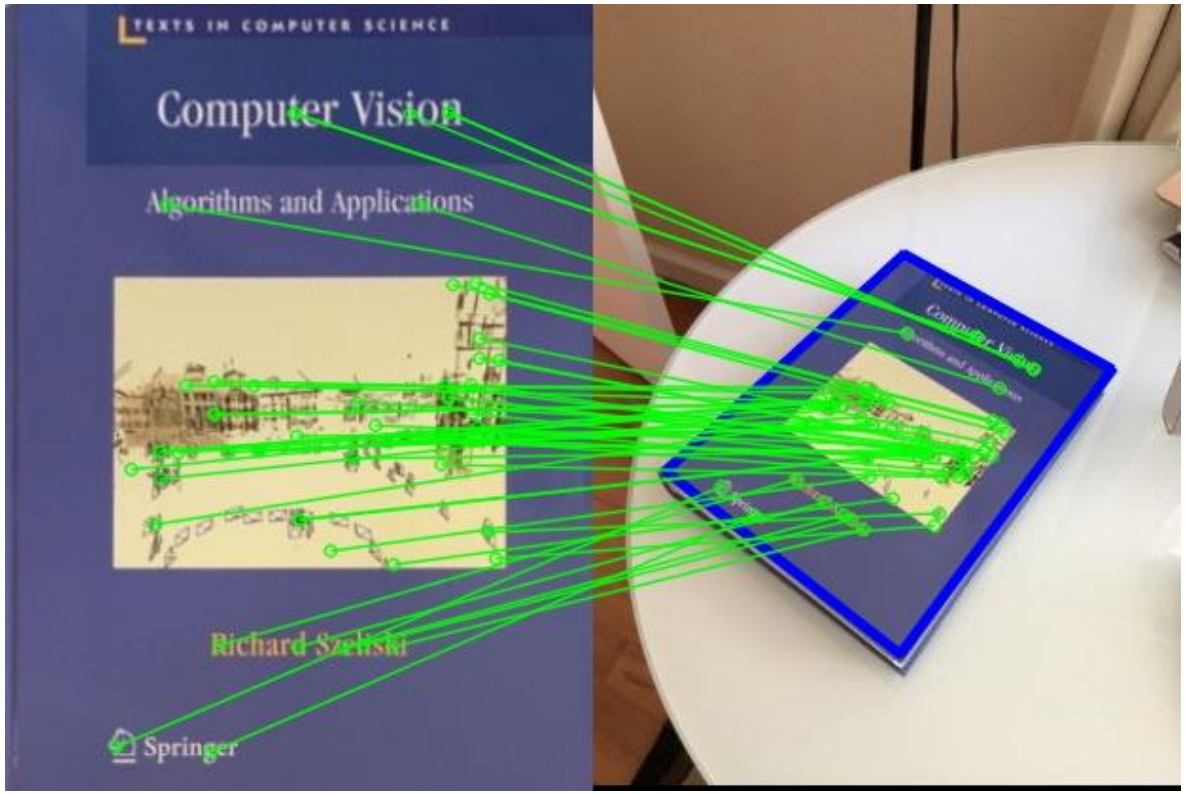


$$\begin{pmatrix} \tilde{x} \\ \tilde{y} \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \Delta(x, y)$$

Tips

Observe: The convention is to

- estimate the transformation <u>from the target to the source image</u>
- transform the source to the target coordinate system

I.e. for each point in the target, find the coordinates where to get pixel values from the source.

- A **minimal solver** needs only K measurements
- Remember the **homogeneous matrix**
- Try to understand the supplied functions.